

Shivashant Manohar

Bengaluru, India | shivashant.work@gmail.com | +91 93538 31815
github.com/hey-shiv | linkedin.com/in/shivashant | hey-shiv.github.io | x.com/hey_shivv

EDUCATION

VIT-AP University

B.Tech in Computer Science - Artificial Intelligence & Machine Learning CGPA: 8.05/10

2023 - 2027

Amaravati, Andhra Pradesh

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++

ML / AI: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face (Transformers, PEFT, TRL), reinforcement learning (REINFORCE, PPO), RAG, FAISS, ReAct agents, DPO/QLoRA

Data / Audio: pandas, NumPy, SciPy, SQLite, librosa (STFT, mel, pYIN), dynamic time warping, pre-registered and matched-control experiments, bootstrap CIs

Engineering: FastAPI, Hono, React, REST APIs, Drizzle ORM, pytest, Playwright, GitHub Actions CI, Vercel, Render, Git, Linux, Google Cloud Platform

PROJECTS

The First Song - Emergent Acoustic Communication between RL Agents | PyTorch, audio DSP

2026

- Built a multi-agent reinforcement-learning system from scratch in PyTorch (REINFORCE with a learned baseline, GAE, PPO) in which senders emit continuous notes (pitch, duration, gap, loudness) through a fixed audio renderer and listeners act only on noisy mel-spectrograms, rewarded solely for coordination.
- Tested whether a shared acoustic channel makes 2 to 4 sender-listener pairs reorganise their calls in frequency or time by comparing against matched private-channel controls with pre-registered paired and equivalence tests, using separate probes for physical source separation and mixture decoding.
- Counted structure as real only when a pre-registered metric beat four baselines, measured with pYIN pitch tracking, onset detection and overlap metrics, and made every condition-by-seed run reproducible through manifests (config hash, seed, commit, package versions) and checkpoints.

Structure-Aware Hybrid Retrieval for Cover Song Identification | PyTorch, NumPy, pytest

2026 - Present

- Cut query time from 2.63 h to 38 ms on the Da-TACOS benchmark by building a two-stage retrieval system: a PyTorch TCN encoder (6 dilated residual blocks, 128-d embeddings, SupCon/triplet loss) shortlists 30 candidates, then a 12-key-rotation subsequence-DTW aligner reranks them.
- Raised hybrid MAP 6.8x (0.018 to 0.122), to within 1.11x of the best system, by running four controlled experiments (hubness correction, 384-frame reranking, training on all 4,780 works, longer training) after the first benchmark showed exhaustive alignment winning 4.6x.

AgentAlign-Lab - Verifier-Guided Preference Learning for LLM Agents | PyTorch, TRL, PEFT, Gradio

2026

- Replaced human preference labels with deterministic verifiers for terminal agents by building a ReAct tool-use loop that runs 60 sandboxed tasks across five families (Python bug-fix, config repair, data transformation, log extraction, safety traps) and scores each trajectory on pass/fail, failed commands and unsafe actions.
- Converted scored trajectories into 336 DPO preference pairs and built a QLoRA training path (4-bit NF4 via bitsandbytes, LoRA via PEFT/TRL) for Qwen2.5-Coder-1.5B, with a base-vs-adaptor evaluation on a held-out test split and a Gradio dashboard for inspecting trajectories.

Coil - NSE Stock Screener and 10-Year Backtest | Python, pandas, SQLite, FastAPI

2025 - 2026

- Showed that a popular chart pattern has no demonstrable edge (11.0% vs 22.3% CAGR for matched random entries, 97 vs 94 trades, costs deducted) by building a ten-year point-in-time backtest that replays 2,657 signals against same-stock, same-risk random controls on a survivorship-controlled top-500 universe.
- Shipped a live web app (coil-vcp.vercel.app) backed by 390+ tests over ~15k lines of Python by serving a FastAPI backend on Render, a vanilla JavaScript frontend on Vercel, and a nightly GitHub Actions job that refreshes prices and publishes a data snapshot.

Frontier - LLM Project Recommendation Engine | TypeScript, React, Hono

2026

- Kept generation available when free-tier APIs failed or hit rate limits by writing a sequential provider chain (Gemini, Groq, OpenRouter, local fallback) that tags every response with the provider that served it.

LEADERSHIP & COMMUNITY

Google Developer Groups VIT-AP - Cloud & DevOps Team

Aug 2024 - Present

Worked with GCP and cloud-native tooling; contributed to developer workshops and community sessions.